GROUP PROJECT PRESENTATION

Loan Approval Prediction

Using Machine Learning

Comparing models that predict loan application outcomes

LOAN
PREDICTION

GROUP PROJECT

Integral University

Data Science Fundamentals

GROUP MEMBERS

SHEIKH MOHAMMAD SAALIM | MD KAIF | ASHER UDDIN

FAIZ AHMAD | GULAM ZAID

Predicting a loan application outcome

PROJECT GOAL

We trained models to predict whether an application in the dataset was approved or not, using details such as income, loan amount, credit history, and property area.

DATASET

The Kaggle Loan Prediction dataset contains applicant details and a Loan_Status column. Y means approved; N means not approved.

WHAT THIS PROJECT CAN SHOW

It shows how classification models learn from past examples. It is a classroom exercise, not a real bank approval system.

The model predicts patterns in this dataset. It cannot decide fairly or reliably for real applicants without much more testing.

How we checked and prepared the data

EXPLORED WITH PANDAS

- Checked rows, columns, and missing values
- Counted approved and not approved applications
- Compared decisions by credit history

PREPARED FOR THE MODELS

- Filled text blanks with the most common answer
- Filled LoanAmount blanks with the middle value
- Changed categories to numbers; removed Loan_ID

We used 80% of the applications to train the models and kept 20% aside to check their predictions. The same test group was used for all three models.

Logistic Regression inputs were scaled; the tree models used the original inputs.

Three models, checked the same way

MODELS TRAINED

- Logistic Regression: finds patterns in the inputs
- Decision Tree: asks a series of yes/no questions
- Random Forest: combines 100 tree predictions

HOW WE COMPARED THEM

Each model was tested on the same applications it had not seen during training. We compared accuracy, precision, recall, and F1-score.

Accuracy: share of all predictions that were right. Precision: how many predicted approvals were actually approved. Recall: how many actual approvals the model found. F1 combines precision and recall.

Precision, recall, and F1-score below are for the Approved class (1).

RESULTS

Model results on the test applications

The test set was held back from training. Scores are rounded to two decimal places.

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	86.18%	84.00%	98.82%	90.81%
Decision Tree	84.55%	82.35%	98.82%	89.84%
Random Forest	82.11%	84.62%	90.59%	87.50%

In this comparison, Logistic Regression had the highest accuracy and F1-score. Random Forest had the highest precision. The best choice depends on which mistakes matter most.

What our Colab demo shows

CURRENT DEMO

- Tests one saved application from the test data
- Shows each model’s predicted outcome
- Keeps the trained models and input columns

LIMITS AND NEXT STEPS

- Small dataset; background details are limited
- Past decisions may include unfair patterns
- Demo runs in Colab; not a real loan service

Before any real use, a model would need larger and more suitable data, careful fairness checks, clear reasons for each decision, and human review.

What we learned

We used Python and Pandas to explore applicant data, fill missing values, prepare the columns, train three classification models, and check their predictions on a test set.

Logistic Regression gave the strongest accuracy and F1-score in our comparison. These results describe this small dataset only; they do not prove that the model is suitable for real loan decisions.

DATASET SOURCE

Loan Predication - Kaggle

https://www.kaggle.com/datasets/ninzaami/loan-predication

PROJECT CODE

https://colab.research.google.com/drive/1lcCpSKfGXhTxwIuo0FzjY6s3KXeOGkk6?usp=sharing